11 Motion Control

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11.1 What is Motion Control?

Motion control of machines and processes began with humans turning cranks or moving levers to actuate motion as they watched a measuring scale. Their brains were the control as they compared the desired position to the actual scale and took whatever corrective action was necessary to bring them into agreement. As automation was introduced, the scale was replaced with a feedback device; the human muscle was replaced with an actuator (motor); and the brain was replaced with electronics (controller). An operator (or input device) would enter the desired position; the controller would compare the feedback position to this desired position and decide the direction and speed needed to achieve the desired position. The controller would send these instructions to the motor on a continual basis until the desired position was achieved. Initially, machine tools were the major beneficiary of this automation. Today, packaging, material handling, food and beverage processing, and any industry that uses machines with movable members are enjoying the benefits of motion control.

11.2 Advantages of Motion Control

Saving time is a major benefit of motion control. It might take a person a minute or two to hand crank a machine a short distance and align it with the scale. A typical servo will do it in less than 0.5 seconds. Accuracy is another plus. In 0.5 seconds, the servo will get the machine to within the accuracy of the system. To achieve this accuracy manually may take verniers or other time-consuming means.

Coordinating two axes is impossible with manual hand cranks, but easy with servos. Servo clamping is another benefit. As will be seen shortly, a servo will return an axis to position when an outside force has moved the axis.

11.3 Feedback

The feedback device can be considered the eyes of the system. It determines the velocity and the position of the axis in a motion system. Motion control systems use many different types of feedback devices, which can be analog or digital and come in both incremental and absolute configurations. Both incremental and absolute types of devices can track position changes: the difference is in how they respond to a loss of power. Absolute devices can determine their position on power up, providing the axis was calibrated during start up. Incremental devices will lose their position and need to go through a homing sequence on power up. In this chapter we will briefly discuss several of the more popular types of feedback devices.

11.3.1 Resolvers

Resolvers are analog devices relying on magnetic coupling to determine position. They do this by looking at the magnetic coupling of a rotating winding, the rotor, compared to two stationary windings, stators (see Figure 11-1). This coupling varies with the angle of the shaft relative to the stators. As such, resolvers are rotary transformers and typically are interfaced with an A/D circuit to be used with a digital controller. These circuits typically provide 12 or 13 bits of resolution, although there are models with up to 16 bits of resolution. Resolvers are commonly used as velocity feedback devices in brushless AC servomotors. They are extremely rugged and provide absolute feedback for one revolution of their shafts. This makes them ideal for AC servomotors, as the absolute nature in one rev allows the drive to know where the motor shaft is. This allows it to commutate the motor. Although this works well for commutation, it is less than ideal for position absolute feedback. Typically an axis travels over more than 1 revolution of a motor shaft; therefore a single resolver loses its ability to act as an absolute device. If the application allows the axis to perform a homing routine on power up, this configuration offers the advantage of using a single feedback device. You can use the resolver for the drive for commutation and the controller for position and velocity. To achieve absolute positioning over multiple turns, use a dual resolver set or master vernier resolver set. Two resolvers are connected to the load, but each resolver is geared at a different ratio. By looking at the phase shift between the two resolvers it is possible to determine the absolute position of the axis over multiple turns.

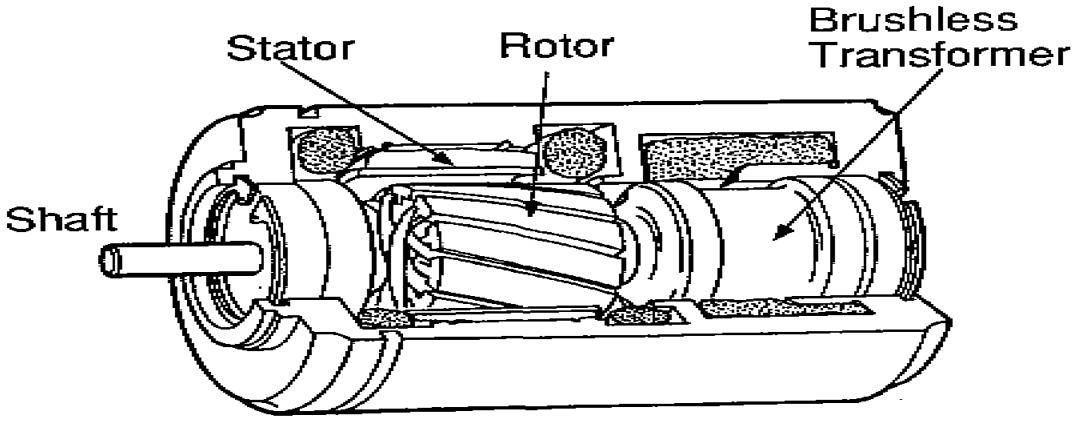


Figure 11-1: Resolver

11.3.2 Magnetorestrictive Transducers

Magnetorestrictive transducers are unique due to the noncontact nature of this type of feedback device, which makes them ideal for linear hydraulic applications. Magnetorestrictive transducers operate in a manner similar to sonar. A sensing magnet is placed on or, in the case of a hydraulic cylinder, inside the actual load. The magnetorestrictive transducer sends out a pulse to the moving magnet, which causes a mechanical strain that is conducted back to the unit. The time it takes for the strain to conduct back determines the position of the axis. These devices are absolute and must be used in linear applications. The downside to these devices is their limited resolution, typically several hundred counts per inch with moderate accuracy.

11.3.3 Encoders

Encoders are extremely popular in motion control applications. They are digital, relatively inexpensive, and have very high resolution and accuracy. Encoders come in both incremental and absolute configurations and are available from many vendors.

Incremental encoders are extremely popular and are used in many motion control applications. They are basically discs with slots cut in them and a through-beam photo sensor (see Figure 11-2). This configuration creates a pulse train as the encoder shaft is turned. The controller counts the pulse train and thereby determines how far the axis has traveled from a known position. This known position is determined at power up by going through a homing sequence. Typically there are two photo detectors, channel A and B, set 90 degrees apart. By looking at which channel rises first, the controller determines the direction of travel. The price of incremental encoders is primarily determined by their resolution. Incremental encoders of 1000 counts or less are very inexpensive. Incremental encoders with more than 10,000 counts are considerably more expensive. Specialty encoders with greater than 50,000 counts per rev are available but are very expensive.

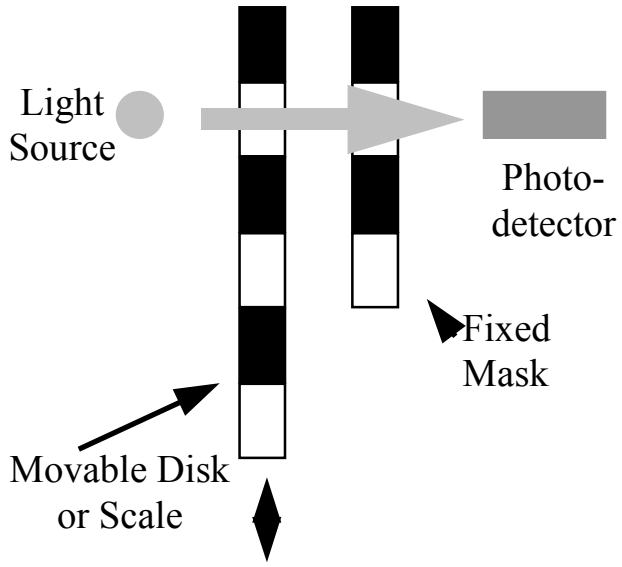


Figure 11-2: Incremental Encoder

Absolute encoders, as their name suggests, are absolute devices. Instead of creating a pulse train, an absolute encoder uses a disc that reveals a specific binary or gray code based on the position of the disc. Absolute encoders can be either single turn or multiturn. A single turn absolute encoder, like a resolver, gives an absolute position over one turn of its shaft. A multiturn absolute encoder incorporates an integrated gear that is encoded so that the number of turns can be recorded. Like incremental

encoders, the higher the resolution, the more expensive the encoder is. Typical systems use absolute encoders with 12 bits (4096 counts) or 13 bits (8192 counts) per revolution. In addition, a multiturn absolute encoder will record 12 or 13 bits of revolutions. Typically a multiturn encoder will be described by the resolution bits plus the revolution bits; therefore a 13-bit (8192) resolution disc with the ability to track 12 bits (4092) revolutions will be referred to as a 25-bit multiturn absolute encoder. Many of today's absolute encoders are programmable and sit on a variety of industrial networks such as Asi, CAN, Devicenet, or Profinet.

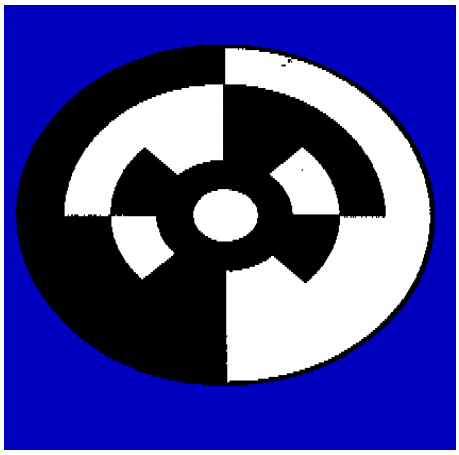


Figure 11-3: A Three-bit Absolute Encoder Rotary Disc

Linear encoders are incremental encoders that have been *rolled* out along the length of an axis. Also known as a glass scale, these encoders have been used for decades in machine tool applications. A reader is placed on the moving axis and picks up the pulse train generated by moving along the glass scale. Today, with the increasing use of linear motors, this form of encoder is seeing more and more use.

11.3.4 Other Feedback Devices

There are many other types of feedback devices that we have not discussed. These include linear variable displacement transducers (LVDTs), laser interferometers, synchros, and even Hall effect devices. These devices, while in use today, are not as prevalent as the previously described types of feedback in industrial systems. LVDTs still see a lot of use in aerospace applications, and laser interferometers are used in very high-precision applications. Today, using special interfaces and encoders with sine and cosine signals, it is possible to achieve 4 million counts per rev! Many servo drive and motor vendors offer this technology, which is patented by a company in Germany.

11.4 Actuators

The actuator takes the command from the controller and moves the axis. Based on the signal coming from the feedback device, the controller will command the actuator to move the axis at a particular velocity until it comes to the desired position. The actuator provides the means of accelerating and decelerating the axis and maintaining its velocity and position. The actuator can be considered the muscles of the motion control system and can be pneumatic, hydraulic, or electric.

11.4.1 Pneumatic

Pneumatic systems employ compressed gas under high pressure to move an axis. Compressed gas is held in a tank and released into an expandable chamber with a rod attached to it. The gas is released into the chamber through the use of an electrically operated valve. As the gas expands in the chamber

it pushes the rod forward. The rod can be pulled back in either through a second chamber on the other side of the rod, or through a mechanical mechanism such as a spring. Due to the compressibility of a gas, pneumatic systems are usually not stiff enough for typical industrial motion control applications. They have limited use in specialty robotics or are used to position point-to-point systems, such as a flipper or diverger. Typically these systems are open loop, relying on a mechanical or prox switch to tell them that the axis is in position.

11.4.2 Hydraulic

Hydraulic systems are used when great force is required to move the axis and its load. Like pneumatic systems, hydraulic systems employ a pump and a valve, but in this case a liquid is used. The liquid is incompressible; thus the system is extremely stiff when tuned correctly. This liquid, hydraulic fluid, is made up of many different chemicals and is typically toxic to people and the environment. Proper equipment maintenance and proper storage and handling are therefore very important, which increases the ongoing costs of the system. However, there are applications that simply must use this actuator due to its capability to provide tremendous force on demand. Typical applications use a magnetorestrictive transducer for feedback and a hydraulic actuator to move large loads. An example is a transfer line in an automotive engine-machining center. The transfer bar, lifting several engine blocks from one station to the next, would be powered by a hydraulic actuator.

11.4.3 Electric

Electric systems consisting of a servomotor and drive are widely used in industry for motion control applications. This combination can consist of permanent magnet DC or brushless AC motors and drives. Today the performance of variable frequency drives and induction motors has increased tremendously. In some simple applications they have been used to control positioning of axis. The demanding high-performance applications still use servomotors over standard drives; however, the two types of drives are evolving together. Today many servo drives can use either synchronous (servo) or standard induction motors. There is also an increase in the use of linear motors. A linear motor is basically a synchronous motor rolled out on a flat plane. Linear motors are capable of incredible acceleration, high force, and very accurate positioning.

11.5 Electric Motors

The motion control industry uses several types of electrical motors. The stepper motor is used in simple applications, because it is less expensive than a servo. It is popular in its simplicity. You simply tell it how many steps you want it to move. Typically, it doesn't have feedback; you assume it moves the programmed amount.

Many motion control applications do not require positioning. A variable frequency drive runs a motor at a speed proportional to the input signal. For turning a drill or milling cutter, this is sufficient. When you need better speed regulation, use velocity feedback to close a velocity loop. The velocity regulation is determined by how good the velocity feedback device is.

More demanding applications use servo drives and motors, which are typically sold as a set from a particular vendor. This allows the motor to be precisely matched to the drive to extract the maximum performance from the system. Permanent magnet DC and brushless AC are the most common types of servomotors.

11.5.1 Permanent Magnet DC

Permanent magnet DC servos are DC motors that use a permanent magnet to create a field rather than a field winding with current as is used in most DC motors. This eliminates the need for a field winding, but increases cost because it requires high-quality rare-earth magnets. DC motors use a wound rotor. The rotor has windings in it to carry current; this current creates a second magnetic field that is acted upon by the field created by the permanent magnet. This produces torque on the motor shaft. To keep

the rotor turning, the current must switch directions every half revolution. A modern DC motor is more complex than this, but this example demonstrates the theory. A DC motor uses brushes to switch current in the windings on the rotor. This is known as a brush commutator.

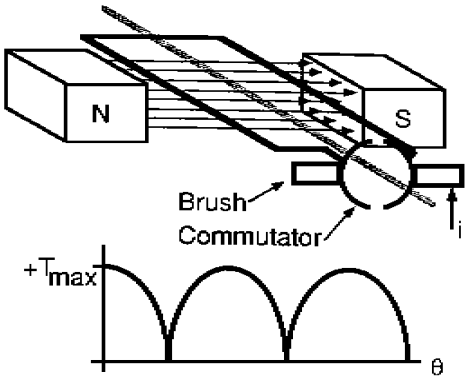


Figure 11-4: Permanent Magnet DC Motor

You must replace the brushes in DC servomotors after extended running time due to abrasion caused by the brushes contacting the moving rotor. The brush commutator also suffers from an effect called arc over. This occurs when the speed of the motor becomes great enough to induce a spark from one plate to another on the commutator. Due to the wound rotor, DC motors tend to have poor heat dissipation. Still, they are relatively inexpensive and require fewer electronics than AC servo motors do.

11.5.2 Brushless AC

Brushless AC motors are a permanent magnet DC motor turned inside out. The windings move to the stator, and the magnets move to the rotor. As the name suggests, the brushes are no longer used. Instead they employ electronic commutation and divide current into the appropriate windings to keep the motor turning. The windings are typically configured in a three-phase Y. Using a resolver on the rotor allows the drive to know where the rotor is, and therefore where the magnets are. Using this information allows the current in the windings to be adjusted to produce smooth torque. The division of the current in the windings produces sinusoidal commutation; thus these motors are referred to as synchronous AC motors. Heat dissipates better with this configuration, because the windings are in the stator and thus are able to use the body of the motor to dissipate heat. This allows for higher peak torques compared to the equivalent DC motor. With the elimination of the brushes there is no arc over; therefore AC motors are able to achieve higher speeds.

11.6 Controllers

The controller is the brain of the system. Its commands can be entered manually or downloaded. Manual entry occurs through the operator's station (keyboard, switches, etc.) The earliest controllers used punched paper tape to enter prepared commands, but almost any computer friendly media is possible now, as is downloading from a host computer and even a Web browser. The controller executes the program, produces the command signal, reads the feedback, compares it to the command, and decides the action that needs to be signaled to the amplifier/motor to bring the difference to zero.

The motion controller has evolved over time. It has gone from a dedicated unit that interfaced with other CPUs on the machine, typically a PLC, to becoming integrated with the PLC. Some vendors have added limited PLC functionality to their motion controllers, while others have integrated the motion controller into the PLC. This has led to many improvements to motion systems, such as eliminating the need to program two CPUs with different languages and to create handshaking routines to deal

with two asynchronous processors. A single software package and single CPU make it easier to program and troubleshoot a modern system. The integrated controllers can be either resident on a PC or in a traditional dedicated hardware package. Either way, today's user will be able to use less space, have fewer points of failure, and use less wiring than the traditional systems of a PLC and dedicated motion controller.

11.7 Servos

A servo is the combination of the three components described above. Its basic block diagram has two elements as shown in Figure 11-5. The summing network that subtracts the feedback number from the digital command, thereby generating an error, is part of the controller.

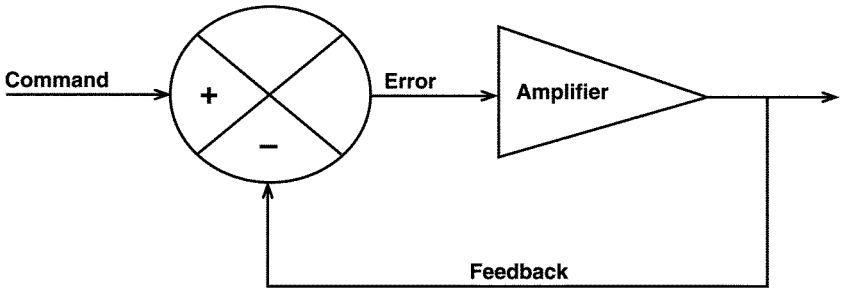


Figure 11-5: Basic Servo Block Diagram

The amplifier consists of the drive and motor. The feedback device that shows the movements is attached to the physical motor output or to the load. The error signal from the summing network to the drive can be either digital or analog, depending on the drive used. The drive is designed to run the motor at a velocity proportional to this error signal.

The beauty of a servo is the error must be zero for the motor to be at rest. This means the motor will continue to drive the load in the proper direction until the command and feedback are equal. Only then will it remain at rest. For positioning servos, this has the added benefit that if some external force disturbs the load, the resulting position loop error will force the motor back into position.

Servo motion control systems typically consist of three cascaded loops. The innermost loop is the current or torque loop. This loop is typically set by the vendor of the drive motor package and controls the amount of torque produced by the motor. Torque is produced by current: the more current in the motor, the more torque is produced. The amount of torque produced is based on the Kt constant of the motor. The Kt constant is a measure in units of torque per ampere. The amount of torque produced by the motor can be calculated by T=Kt*A, where *T* is torque and *A* is the amps going to the motor. This loop always resides in the drive.

The velocity loop is the second loop. This loop takes a velocity command and feeds it into the torque loop to control acceleration or deceleration. This loop can reside in either the drive or the motion controller depending on what mode you are operating the drive in. If the drive is set to torque mode, then the velocity loop resides in the controller. This means the controller will give the drive a torque command. If the drive is in velocity mode, then the drive handles the velocity loop. In this configuration the controller produces a velocity command for the drive. The velocity loop must be tuned by the user, and it determines the ability of the system to accurately follow the velocity command and to overcome disturbances to the velocity of the axis. The velocity loop must always be properly tuned before the user attempts to tune the position loop.

The position loop is the outermost loop, and it always resides in the motion controller. This loop is the final loop the application engineer must tune. The methods employed to tune a position loop are similar to the methods used to tune any other loop. In the end the user typically verifies performance by looking at following error. Following error is the difference between the command and actual position. Typically following error will jump when an axis accelerates or decelerates, and it tends to become smaller when the axis is at steady speed. By using velocity feedforward, you can minimize the jump in position error when changing speeds. The smaller the following error is at speed, the hotter the axis is tuned. Depending on the application this can be good or bad. In a point-to-point move an extremely hot system will tend to overshoot, which would not be desirable. For an application requiring two axes to be synchronized, an electronic gearing or camming situation, it is desirable to minimize following error. It is up to the user to determine the best method of tuning the axis and what level of performance is expected.

11.8 Feedback Placement

When the feedback is directly on the motor shaft (the usual case), the shaft is exactly where the servo is being commanded to go when the system comes to rest. Because there may be a coupling, gearbox, rotary-to-linear converter, or other element between the motor and the actual load, the load may not be perfectly in position. Backlash and wind-up are the two main culprits explaining the discrepancy, so the mechanical design has to keep these below the maximum error allowed. An alternative is to place the feedback on the load itself, but now the backlash and wind-up are within the servo loop and may present stability problems. There are notch filters and other compensation networks that can alleviate some of these stability problems, but they are not for the novice. A solid mechanical design is paramount.

11.9 Multiple Axes

Most applications require multiple axes, so the coordination of those axes becomes a major issue. Most "set-up" type applications allow the axes to run independently and reach their positions at different times. For instance, corrugated box making requires that dozens of axes be moved during setup to accommodate the different box sizes. These axes will determine where the cuts, creases, and printing occur. They can all be moved independently. However, once the machine starts running the boxes, any movements must be coordinated. Historically, gears and cams coordinated these motions mechanically, but electronics is taking over many functions.

11.10 Leader/Follower

Many machines (automotive transfer lines or cereal packaging) were originally designed with a line shaft that ran the length of the machine. Each station where particular operations occurred had one or more cams attached to the shaft to synchronize the operations within the station and with other stations. The shape of the cam dictated the movement that any particular axis had to make. The entire line was controlled by changing the speed of the line shaft. The electronic leader/follower is replacing these mechanisms. Each axis now has a servo, and its command is generated from a data table. The data table for each axis contains the position needed for that axis at each count position of the leader. The leader could be a counter that allows you to vary the count rate to simulate the line speed changing. This "electronic cam" arrangement allows many axes to be synchronized to a variable leader.

11.11 Interpolation

Linear/circular interpolation is a common way to coordinate two axes. It uses an algorithm that meters out movement increment commands to each axis servo on a fixed-time basis (1 millisecond is a typical time basis). Interpolation was incorporated in the earliest numerical controls for machine tools in the 1950s and is still in common use today. Parts that needed to be milled could get quite complex,

but the theory was that any complex shape could be described with a series of straight lines and circles. The Electronic Industry Association developed a programming standard (RS-274) for instructing the machines in the early 1960s. The linear/circular algorithm executed those instructions.

For linear interpolation, it is only necessary to program the end point and velocity. The controller figures out how far each axis must move in every time increment to get it from its present location to the end point at the required speed. For circles, you must program the center of the circle from the present machine position, the end point on the circle, the velocity, and whether it should execute the circle in a clockwise or counterclockwise direction. The velocity can be changed by operator intervention without affecting the coordination, so the process can be slowed down or sped up as needed.

11.12 Performance

The performance of a motion control axis is typically specified in bandwidth or in response to a step input. Bandwidth refers to the frequency at which the servo loop output begins to fall off from its command. If you command a position loop servo with an AC function, the output will follow exactly at low frequencies. As you increase the frequency, a point is reached where the output begins to fall off, and it will do so rapidly with further frequency increases. The bandwidth is defined as the point where the output is 0.7071 of the input in amplitude. With typical industrial machinery, this is about 3 Hertz. Amplifier/drive builders also use bandwidth to rate the performance of their products. This is the bandwidth of the velocity loop that they provide, not the position loop. When tied to a real machine, you can expect this velocity loop bandwidth to be in the 30-Hertz ballpark. Many vendors might claim 100 Hertz or more, but that is not possible on industrial machinery.

Response to a step input is a measure of how fast the servo will get to its final state when a small command is initiated. You might recall that a spring/mass system gets to 63.6% of its final value in one time constant and follows the natural logarithmic curve. This time constant and the amount of overshoot (if any) are the values to consider for performance. A typical position loop servo on industrial machinery will have a time constant of 50 milliseconds. The amplifier/drive (velocity loop) will have about a 5-millisecond time constant.

11.13 Conclusion

Motion control has offered many benefits in automating factories. Its use continues to expand faster than the economy grows, as companies convert many applications to this better technology. The vendors are also making controllers easier to apply and tune, thereby taking some of the previous mystique away.

11.14 References

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About the Authors

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Lee Lane has a BSEE from the University of Maine and has over 14 years of experience in Industrial Automation. He has moved into positions of increasing responsibility at Rockwell Automation where he started as an application engineer. Today Lee is a marketing manager in the Logix / Netlinx / Kinetix business of Rockwell Automation. Lee is an ISA member, a Certified Automation Professional (CAP), and is part of the CAP steering team. Lee has been very involved with the formation of the first CAP tests, the study guide, and in promoting CAP.